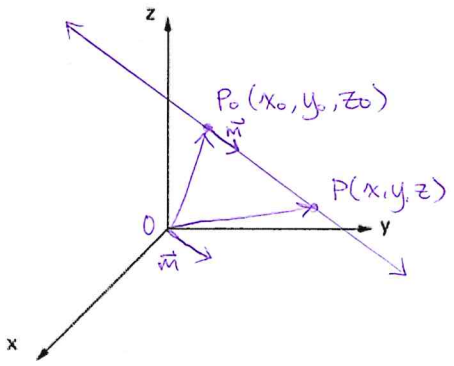
Equation of a Line in R^3

Similar to the derivation in R^2



Let $P_0(x_0, y_0, z_0)$ be a known point on the line.
Let $\vec{m} = (a, b, c)$ be the direction vector.
Let $P(x, y, z)$ be any point on the line.

$\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P}$ but $\overrightarrow{P_0P} \parallel \vec{m}$

$= \overrightarrow{OP_0} + t\vec{m}$, $t \in R$ $\therefore \overrightarrow{P_0P} = t\vec{m}$, $t \in R$

Rename $\overrightarrow{OP} = \vec{r}$

$\overrightarrow{OP_0} = \vec{r_0}$

Vector Equation in R^3

$$\vec{r} = \vec{r_0} + t\vec{m}, \quad t \in R$$

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c), \quad t \in R$$

Parametric Equations in R^3

$$\left.\begin{aligned} x &= x_0 + ta \\ y &= y_0 + tb \\ z &= z_0 + tc \end{aligned}\right\} \; t \in R$$

Symmetric Equations in R^3

$$t = \frac{x - x_0}{a} \qquad t = \frac{y - y_0}{b} \qquad t = \frac{z - z_0}{c}$$

$$\therefore \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}, \quad \text{for } \vec{m} = (a, b, c) \text{ where } a, b, c \neq 0.$$

Note: A line in R^3 does not have a single unique NORMAL VECTOR.
Therefore, a line in R^3 does not have a CARTESIAN equation.

Example 1: Determine the vector, parametric and symmetric equations of a line passing through $P(1,-4,3)$ and $Q(2,-1,-5)$.

$\vec{m} = \overrightarrow{PQ} = (1, 3, -8)$

Vector:

$\therefore \vec{r_1} = (1, -4, 3) + t(1, 3, -8), \quad t \in R$

OR $\vec{r_2} = (1, -4, 3) + s(2, 6, -16), \quad s \in R$

OR $\vec{r_3} = (2, -1, -5) + k(1, 3, -8), \quad k \in R$

Parametric:

$$\left.\begin{aligned} x &= 1 + t \\ y &= -4 + 3t \\ z &= 3 - 8t \end{aligned}\right\} \; t \in R$$

Symmetric:

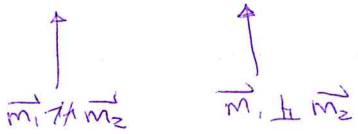
$$\frac{x - 1}{1} = \frac{y + 4}{3} = \frac{z - 3}{-8}$$

$$x - 1 = \frac{y + 4}{3} = -\frac{(z - 3)}{8}$$

$\vec{m}_1 \not\parallel \vec{m}_2$ and share one point.

Example 2: Classify the following two lines as equivalent, parallel, perpendicular, or none of the above.

$\vec{r}_1 = (1,-4,3) + t(1,3,-8)\,,\ t \in \Re$
$\vec{r}_2 = (2,-1,-5) + s(-2,-6,16)\,,\ s \in \Re$



Compare directions

$\vec{m}_1 = (1,3,-8)$ $\vec{m}_2 = (-2,-6,16)$

① Check parallel:

Solve for k. (if parallel)

$(1,3,-8) = k(-2,-6,16)$

$1 = -2k$ $3 = -6k$ $-8 = 16k$

$k = \frac{-1}{2}$ $k = \frac{-1}{2}$ $k = \frac{-1}{2}$

$\therefore$ parallel.

② Check Equivalent

Check if point $(2,-1,-5)$ lies in L_1.

$2 = 1+t$ $-1 = -4+3t$ $-5 = 3-8t$

$1 = t$ $3 = 3t$ $-8 = -8t$

$1 = t$ $1 = t$

$\therefore$ same value t; point $(2,-1,-5)$ is on line 1.

$\therefore$ L_1 is equivalent to L_2.

Note: check $\perp$.
$\vec{m}_1 \cdot \vec{m}_2 = 0$.
If not parallel or perpendicular, it would be "neither" for now.

Example 3: Determine the symmetric equations of a line that goes through the point $(3,-1,2)$ in the direction of $\vec{m} = (4,0,-5)$.

$\vec{r} = (3,-1,2) + t(4,0,-5)\,,\ t \in \mathbb{R}$

$\left.\begin{array}{l} x = 3+4t \\ y = -1 \\ z = 2-5t \end{array}\right\}\ t \in \mathbb{R}$

$\therefore\ t = \frac{x-3}{4}$ $t = \frac{-(z-2)}{5}$ and $y = -1$

we write as a combination: $\frac{x-3}{4} = \frac{-(z-2)}{5}$, $y = -1$

$\frac{x-3}{4} = \frac{-(z-2)}{5}$: symmetric

$y = -1$: parametric. ← y never changes regardless of parameter t.